

Beyond the Fixed Load: A Mathematical Model of Strength Adaptation to Accommodating Resistance

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1 Introduction

Optimal strength development is critical not only for athletic performance, but also for injury prevention, rehabilitation, and long-term functional health. Musculoskeletal injuries accounted for an estimated 214 billion USD in direct medical costs and lost productivity in the United States alone between 2012-2014 and have likely only increased since then [5]. Strength deficits are independently associated with increased fall risk in older adults, with falls representing the leading cause of injury-related death in people over 65 [2]. However, quantifying strength progression remains a challenging task, because human adaptation to resistance training – the process by which muscles become stronger, more efficient, and more fatigue-resistant as a result of repeated weight training – unfolds on different time scales and varies significantly between individuals. Consequently, predicting how an athlete’s strength will develop under a given training program is difficult. Because of these complexities, constructing a mathematical model of strength adaptation offers substantial practical and theoretical benefits. For example, such a model could help coaches optimize strength gain and injury prevention in their weight training, allow sports medicine clinicians to design more effective rehabilitation programs, and generate qualitative predictions about long-term strength outcomes that would be too slow or expensive to obtain through controlled experiments on real athletes.

A key obstacle to modeling strength adaptation is that muscle force production on a weighted load is not a single number but a function of the position of the load and the speed at which it is moving. The amount of force a muscle can generate depends on how stretched or compressed it is, as muscles produce peak force near their resting length and progressively less as they shorten or lengthen beyond that point, contributing to the position portion of the function [4]. Speed is taken into account from the muscular pattern of less force produced the faster a muscle contracts, a relationship described empirically by Hill (1938) and a cornerstone of subsequent models on muscle mechanics [4]. In a compound exercise such as the bench press, these two effects combine across multiple muscle groups simultaneously,

producing a variation in the force an athlete can apply throughout the range of motion of the exercise. From this, it is clear that athletes do not have uniform strength throughout a lift, but instead particular weak points across positions and speeds that often determine where and why a maximal lift fails.

Arandjelović (2010) addressed this by constructing a model built from three components chosen to capture the minimum necessary physiology without requiring impractical quantities that are difficult to measure.

1. A **capability profile** - a two-dimensional surface describing the maximum force a muscle can produce at each position and speed. This treats the athlete and their biomechanics as a single system rather than requiring measurements of individual muscles.
2. A **force-motion equation** - a differential equation derived from Newton’s second law that governs how the load accelerates given the force the athlete applies. This equation helps distinguish bands, chains, and fixed loads as different forms of resistance, as each produces a different governing equation.
3. An **adaptation update** - a rule based on the principle of specificity, which states that the capability profile is updated after each training session in proportion to how closely the position and velocity conditions experienced during the lift match each region of the profile. In other words, strength gains are specific to the positions and velocities trained, rather than transferring uniformly across the range of motion. [1]

We implement this model computationally to compare how three loading strategies – fixed load, elastic bands, and heavy chains – affect long-term strength development in the same hypothetical athlete. Specifically, we generate an initial capability profile, simulate the fatigue accumulation across a six-repetition set, and run 100 repeated training sessions of the bench press under each loading condition, tracking how the predicted one-repetition maximum (1RM), or heaviest load an athlete can lift exactly once, evolves over time. This allows us to directly compare the long-term adaptive consequences of each training method under identical conditions.

2 Model and Methods

2.1 Capability Profile

To model an athlete’s strength throughout a lift, we represent their force-producing capacity using a capability profile $\hat{F}(x, \dot{x})$ following the approach of Arandjelović (2010). The capability profile describes the maximum force in Newtons that an athlete can exert against a weighted load as a function of two variables, x representing the displacement of the load in meters, measured upward from the bottom of the lift, and $\dot{x}$ signifying $\frac{dx}{dt}$ or the velocity of the load in meters per second. We use the term capability profile throughout to refer to the two dimensional surface such as that shown in Figure 1, which is determined from the

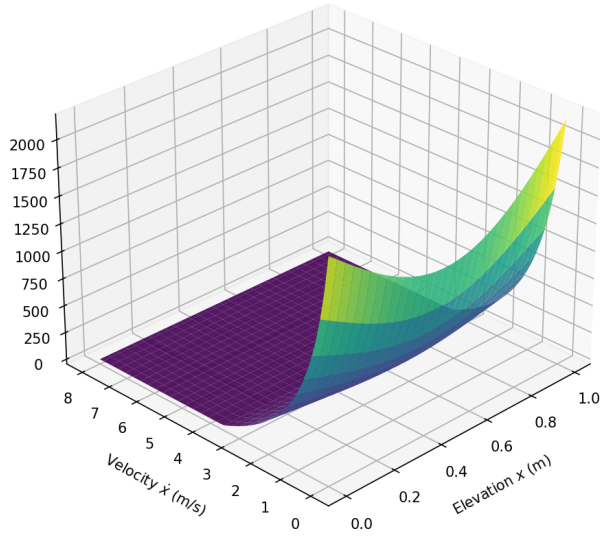


Figure 1: The initial capability profile $\hat{F}(x, \dot{x})$ (Equation 3) visualized over the elevation-velocity plane. Force peaks at low velocity and mid-to-high elevation, decaying to zero as velocity approaches the maximum concentric contraction speed. The z-axis, though not labeled above, represents force in Newtons.

same mathematical expression in all contexts – the capability function. Generally, there is no distinction between these two terms, and they are interchangeable.

The capability profile draws on the well-known relationships between force and length/velocity: elongation and contraction. When a muscle is stretched or shortened beyond its optimal length, maximum force production decreases [4]. Therefore, the position of the weighted load in the lift directly affects the force an athlete can exert. Similarly, an increase in the rate of change of muscle length, also known as contraction velocity, decreases maximal force production [4]. Together, these two dynamics cause the capability profile to form a two-dimensional surface, capturing how an athlete’s strength varies throughout a lift.

Working against the maximum force an athlete can exert (the capability function), fatigue reduces F , the total force produced against the load at a discrete moment in the lift. In other words, the capability profile is the maximum force an athlete could possibly produce at a given position under ideal conditions, while F is the applied force actually produced at time t in a given lift, a quantity diminished by accumulated fatigue. Arandjelović separates this effect multiplicatively:

$$F(x, \dot{x}, t) = \hat{F}(x, \dot{x}) \cdot \tilde{F}(t) \quad (1)$$

where t denotes the time elapsed since the beginning of the lift in seconds, and $\tilde{F}(t)$ is the fatigue multiplier. As fatigue builds from sustained muscular effort, the athlete’s available

force output decays exponentially, giving

$$\tilde{F}(t) = e^{-t/\tau_f} \quad (2)$$

where τ_f is a fatigue time constant in seconds. This is a solution to the ordinary differential equation $d\tilde{F}/dt = -\tilde{F}/\tau_f$, which models fatigue as a process of continuous exponential decay in force output. Arandjelović articulates that $\tau_f = 65$ seconds was calculated so that the simulated athlete fails on the sixth repetition when using their five-repetition maximum load of 100kg, consistent with the experimental setup of the paper.

The setup of the initial capability profile takes the form

$$\hat{F}_{\text{init}}(x, \dot{x}) = 2180 \cdot \underbrace{\frac{1}{3}(3 - \cos((x - 0.5)\pi))(0.9 + 0.1x))}_{\text{elevation modulation}} \cdot \underbrace{0.125 \cdot H\left(\frac{10}{1 + 1.125\dot{x}} - 2\right)}_{\text{velocity modulation}} \quad (3)$$

where $H(\cdot)$ a gating function that clamps negative values to zero, enforcing the constraint that force drops to zero at maximum concentric velocity rather than becoming negative. The elevation modulation term reflects a dip in force around the middle of the lift, reflecting a common biomechanical decrease in load-based lifts such as the bench press, where the bottom and top of the lift are more favorable positions for the muscle regions most active in the lift. The velocity modulation term enforces Hill's force-velocity relationship – force decreases as velocity ($\dot{x}$) increases, reaching zero at the maximum possible speed of concentric muscle contraction. The scalar 2180 Newtons of force establishes an overall strength level, and was chosen to reflect an initial one-repetition maximum (1RM) of 125 kg.

2.2 Equations of Motion

During the lift, Newton's second law is applied to the acceleration ($\ddot{x} = \frac{d^2x}{dt^2}$) of the barbell. The exact equation changes depending on the form of resistance used, and the difference in this equation is the main basis of comparison for the following three loading modalities.

2.2.1 Fixed Load

For a barbell of mass m measured in kilograms with no accommodating resistance, the equation of motion is

$$m\ddot{x} = F - mg \quad (4)$$

which gives the following expression for acceleration:

$$\ddot{x} = \frac{F}{m} - g \quad (5)$$

Note that here $g = 9.81 \text{ m/s}^2$ represents gravitational acceleration and $F = F(x, \dot{x}, t)$ is the athlete's applied force, the product of the capability profile and the fatigue multiplier.

2.2.2 Elastic Bands

Elastic bands exert a restoring force kx proportional to displacement, where k is the elastic constant of the band in Newtons per meter. Thus, the equation of motion becomes

$$m\ddot{x} = F - mg - kx \quad (6)$$

which gives the following expression for acceleration:

$$\ddot{x} = \frac{F - kx}{m} - g \quad (7)$$

The band resistance is purely position dependent, containing no velocity term.

2.2.3 Chains

Chains behave differently than bands, increasing the effective mass of the system as the load's position increases. In addition, each chain segment lifted from rest as this position increases must be accelerated to the current bar velocity. Because of this nuance, the equation of motion is derived from conservation of energy rather than simple force balance:

$$\ddot{x} = \frac{F - \frac{1}{2}\omega\dot{x}^2}{m + \omega x} - g \quad (8)$$

where ω is the chain's linear mass density in kg/m. The term $\frac{1}{2}\omega\dot{x}^2$ is a velocity-dependent drag, which grows with the square of bar speed and with chain density. The differences in the equations of motion for bands and chains establishes that the two are fundamentally different during dynamic lifting even when configured to produce the same static resistance at every position.

2.3 Numerical Integration

We solve the equations of motion numerically using finite difference approximations with discrete timestep $\Delta t = 0.01$ s, following Arandjelović (2010). Defining state variables $x_j = x(j\Delta t)$ and $\dot{x}_j = \dot{x}(j\Delta t)$, we step the system forward as

$$\dot{x}_{j+1} = \dot{x}_j + \ddot{x}_j \cdot \Delta t \quad (9)$$

$$x_{j+1} = x_j + \frac{\dot{x}_{j+1} + \dot{x}_j}{2} \cdot \Delta t \quad (10)$$

with initial conditions $x_0 = 0, \dot{x}_0 = 0$. A repetition is recorded as successful when $x_j \geq 1.0$ m and as failed when $\dot{x}_{j+1} \leq 0$ before that point. We simulate sets of up to six repetitions consecutively, with the fatigue multiplier $\tilde{F}(t)$ updating continuously between repetitions.

2.4 Adaptation Rule

Following Arandjelović (2010), we update the capability profile after each lifting session according to the principle of specificity, which states that the adaptation is greatest at positions and velocities similar to those the athlete actually experienced during the lift [1]. Let $\mathbf{x}_j = (x_j, \dot{x}_j)$ denote the load's state at timestep j during a lifting session. Then, the change in capability profile at any point $(x, \dot{x})$ in the capability plane is

$$\Delta \hat{F}(x, \dot{x}) = K_d [\hat{F}_0(x, \dot{x}) - \hat{F}(x, \dot{x})] + K_a [\hat{F}_\infty(x, \dot{x}) - \hat{F}(x, \dot{x})] \sum_j G(\mathbf{x} - \mathbf{x}_j \mid \sigma_L, \sigma_V) \quad (11)$$

where $\hat{F}_0(x, \dot{x}) = 0.5 \cdot \hat{F}_{\text{init}}(x, \dot{x})$ represents a capability profile that acts as a baseline without training, $\hat{F}_\infty = 2.0 \cdot \hat{F}_{\text{init}}(x, \dot{x})$ is the athlete's genetic ceiling, $K_a = 0.0015$ is the adaptation rate constant and $K_d = 0.002$ is the detraining rate constant, signifying a loss of adaptations due to time off from training. The Gaussian kernel

$$G(\mathbf{x} - \mathbf{x}_j \mid \sigma_L, \sigma_V) = \exp \left\{ -\frac{(x - x_j)^2}{2\sigma_L} - \frac{(\dot{x} - \dot{x}_j)^2}{2\sigma_V} \right\} \quad (12)$$

enforces the principle of specificity, scaling each point in the capability plane by determining (with elevation and velocity thresholds σ_L, σ_V) how close those points are to each position and velocity state ($\mathbf{x}_j$ that the load passed through during the lift. In our implementation we use $\sigma_L = 0.02 \text{ m}^2$ and $\sigma_V = 0.06 \text{ (m/s)}^2$, following the values specified in Arandjelović (2010). The detraining component of the equation, slowly returning the capability profile to the untrained baseline, is in contention with the adaptation component, driving the profile toward the genetic ceiling only in the regions of the capability plane visited during training.

We quantify the total adaptive stimulus produced by a lifting session as the integral of the positive adaptation over the full capability plane:

$$A = \iint \Delta \hat{F}(l, v) \, dl \, dv, \quad (13)$$

where l and v are dummy variables representing displacement and velocity respectively. In our implementation this integral is approximated numerically as a discrete sum over the 60×60 capability plane grid, multiplied by the grid spacings Δl and Δv . Following the specification of Arandjelović (2010) that the load increments of progressive overloading scale linearly with A and initial increment of 1%, we implement this as $\Delta m = 0.01 \cdot m_0 \cdot (A/A_0)$ where m_0 is the initial base load and A_0 is the total adaptational stimulus from the first training session.

2.5 Software

All simulations were implemented in Python 3 using NumPy for array operations. For bilinear interpolation of the capability profile at grid points falling between the position

and velocity values of the discrete timestep, we used SciPy’s `RegularGridInterpolator`. The capability plane is represented by a 60×60 uniform grid over $x \in [0, 1]$ meters and $\dot{x} \in [0, 8]$ meters per second. The 1RM is calculated at specific training session checkpoints, as computing it is computationally expensive, involving a binary search over the load mass and testing each candidate with a fresh single-repetition simulation ($t_{\text{start}} = 0$, no prior fatigue) and converging to within 0.5 kilograms after 20 iterations. Figures were produced using Matplotlib.

3 Results

Before comparing loading strategies, we had to first confirm that our simulation reproduced known lifting behavior consistent with Arandjelović (2010). Figure 2 confirms two expected properties. In the left panel, lighter loads (90–110 kg) complete the full range of motion with increasing bar velocity throughout, consistent with the net upward force $F - mg$ remaining positive across the entire lift. As load approaches the 1RM of 125 kg, the net upward force narrows to near zero at the top of the lift, producing the characteristic flattening of the elevation curve visible at $t \approx 9.45$ s. Loads of 127 kg and 130 kg stall before reaching 1.0 m, confirming 125 kg as the 1RM boundary.

The right panel illustrates how fatigue accumulates across a set: each successive repetition takes longer as the athlete’s available force output decays, and eventually the sixth repetition fails entirely before reaching the top of the lift. Repetition 1 completes in approximately 3 s, while repetition 5 requires nearly 10 s, reflecting the exponential decay in $\tilde{F}(t) = e^{-t/\tau_f}$. The sixth repetition fails before reaching 1.0 m because the accumulated fatigue has reduced $\tilde{F}$ sufficiently that the athlete’s effective force ceiling falls below the 100 kg load requirement at the top of the lift. This is consistent with $\tau_f = 65$ s, which was calibrated so that failure occurs precisely on the sixth repetition at the 5-repetition maximum load (consistent with Figure 12 of Arandjelović (2010)). With both checks passed, we proceed to the main comparison.

With the simulation validated, we investigated whether the loading strategy significantly affects long-term strength development and whether differences emerge only after prolonged training. We simulated 100 sessions because the effects of accommodating resistance are predicted to accumulate gradually under progressive overload, making short time horizons insufficient to reveal the divergence between our modalities. All three conditions begin at the same baseline 1RM of approximately 125 kg. Bands produce the greatest long-term gain (+23.8 kg), followed by chains (+10.4 kg), with fixed load showing the smallest improvement (+9.7 kg). Each condition follows the same qualitative pattern: rapid initial gains over the first 20 sessions as the adaptation term $K_a[\hat{F}_\infty - \hat{F}]$ dominates, followed by a plateau and gradual decline as the initial training stimulus diminishes and the detraining term $K_d[\hat{F}_0 - \hat{F}]$ increasingly competes. The right panel of Figure 3 expresses these gains relative to baseline, making it clear that accommodating resistance trains a wider range of positions and velocities per session, which drives greater strength gains over time than fixed load alone.

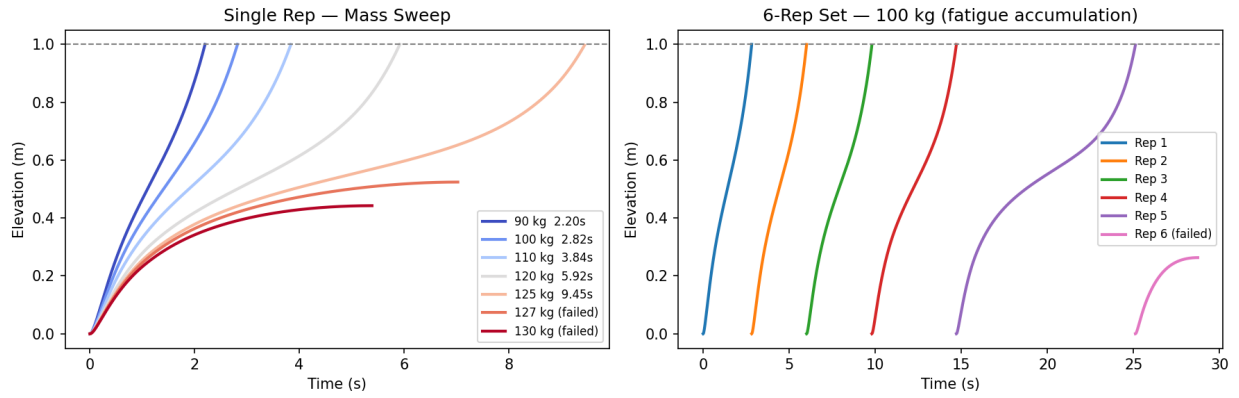


Figure 2: Left: Single-repetition simulations across a range of loads (90–130 kg). Right: Six consecutive repetitions at 100 kg with no rest between reps.

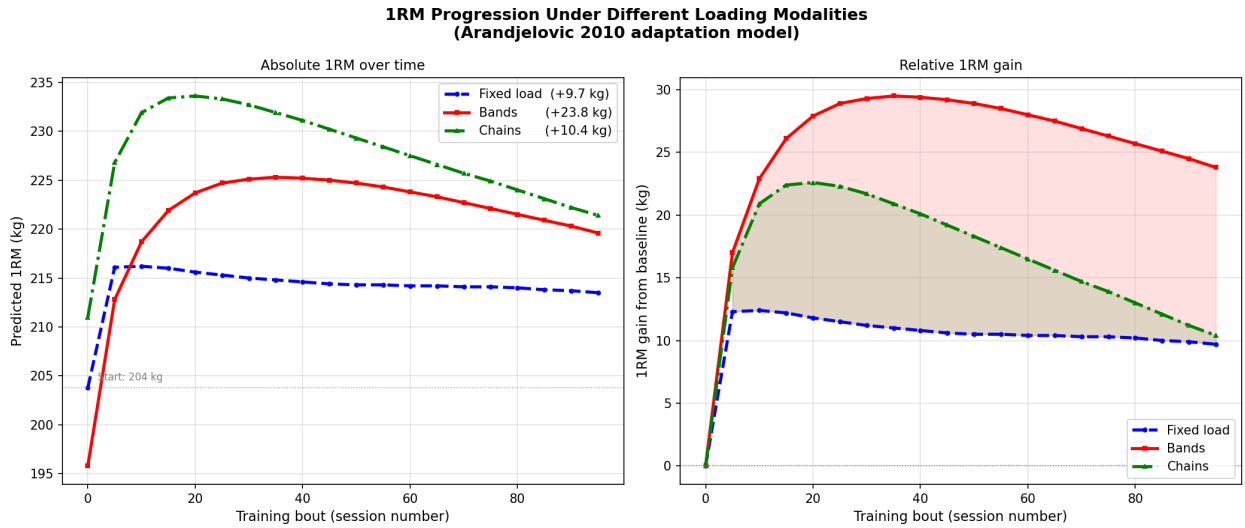


Figure 3: Predicted 1RM (left) and gain from baseline (right) over 100 training sessions under fixed load, bands, and chains with progressive overload. Accommodating resistance produces greater long-term strength gains than fixed load alone.

4 Discussion

Our simulations implement and apply the model of Arandjelović (2010) to investigate how three loading modalities – fixed load, elastic bands, and heavy chains – produce different long-term strength outcomes when applied to the same hypothetical athlete over 100 training sessions. The central finding is consistent with the predictions of the paper, as both forms of resistance outperform fixed loading for long-term 1RM development. These results affirm the motivation that the practical design of resistance training programs would benefit from quantitative modeling rather than trial and error. In particular, the advantage of accommodating resistance is invisible over a short period of training, which suggests that coaches relying on short-term empirical comparisons would underestimate its long-term benefit.

The model structure itself makes this motivation clear. Fixed loading sends the load through a consistent arc in the capability plane every training session. As the athlete adapts, the training stimulus A decays rapidly because the same positions and velocities are being trained repeatedly. These saturated regions reach a plateau as they near the athlete’s genetic ceiling. Accommodating resistance, on the other hand, modifies the load’s trajectory every training session, adapting regions of the capability plane that a fixed load never stresses with the same intensity. Bands in particular keep the load moving through higher-elevation, moderate-velocity regions as progressive overload accumulates. This is one reason why short-term empirical studies show little difference between fixed and accommodating loads, as the effects are only apparent over many sessions of progressive overloading, which most controlled studies are too short to capture.

4.1 Divergent Results

The gap between bands and chains in our simulation is larger than Arandjelović predicts. Arandjelović finds that bands and chains produce nearly identical final 1RM improvements, differing by about 1%. In our simulation, the chains significantly underperform the bands, which is partly due to physical differences between the two and partly due to a choice of implementation. Physically, chains introduce a velocity-dependent drag term $\frac{1}{2}\omega\dot{x}^2$ that has no analogous term in the band equation of motion. This term penalizes faster bar movement and compresses the load’s trajectory toward lower velocities in the capability plane over time. The implementation change arises in how ω – the chain’s linear mass density – is updated under progressive overload. Arandjelović’s paper states that progressive increments are distributed proportionally between fixed and variable resistances but does not specify the exact formula. Our implementation allows ω to grow over the course of training, which amplifies the drag term progressively and increasingly suppresses bar velocity in later training sessions. In practicality, it is likely that athletes do not have access to a wide variety of chains with different densities in this way, and would keep the same chain but increase the weight of the fixed load the chain is attached to. Future simulations holding ω constant may reflect more realistic lifting practices and allow the chain result to converge closer to bands. These results also suggest that athletes should not increase chain density as a part of their

progressive overload in order to achieve greater long term strength growth.

4.2 Assumptions and Approximations

Several simplifying assumptions underlie the model. First, the capability profile $\hat{F}_{\text{init}}$ is defined with a fixed functional form that does not change across the untrained baseline $\hat{F}_0$ and genetic ceiling $\hat{F}_\infty$, both of which are scalar multiples of $\hat{F}_{\text{init}}$. This means that the ceiling is uniform – every region of the capability plane can double in strength. In reality, an athlete with a specific positional weakness might have much more room to grow in that region than elsewhere, meaning $\hat{F}_\infty$ should be shaped differently from $\hat{F}_{\text{init}}$.

Second, the fatigue model $\tilde{F}(t) = e^{-t/\tau_f}$ treats fatigue as an exponential process with constant $\tau_f = 65\text{s}$, assuming a universal decay rate independent of the exercise being performed. This contradicts the principle of task dependency described by Enoka and Duchateau, who argue that there is no global mechanism responsible for muscle fatigue and that the dominant impairment is specific to the stresses of the particular task [3]. One potential reason for this simplified fatigue model is that more complex models require individual athlete and task data that would be difficult to generalize. Additionally, the multiplicative separation of fatigue from the capability profile to give total force produced, $F(x, \dot{x}, t) = \hat{F}(x, \dot{x}) \cdot \tilde{F}(t)$, carries an assumption in itself that fatigue scales the entire capability profile uniformly. That is, this equation assumes that force is reduced by the same proportion at every position and velocity. In reality, fatigue may affect certain regions of the capability plane more than others.

Finally, the model operates on the capability profile of a single exercise in isolation. The model cannot predict the transfer of strength gains across multiple exercises. For example, the strength an athlete gains from a back squat would help with deadlifting, but this relationship is not captured in the model. Incorporating this transfer would require modeling adaptation at the level of individual muscles rather than at the level of the combined exercise output. Arandjelović (2010) also mentions this limitation as a direction for future work, noting that a multi-muscle framework would be necessary to capture carryover between exercises.

4.3 Strengths and Future Discussions

Despite the above limitations, the model has several key strengths. It produces physically interpretable predictions grounded in the physiology of Hill’s force-velocity relationship and the principle of specificity. The feedback loop from capability profile to equations of motion to adaptation back to capability profile is a transparent, mechanistically grounded process that generates predictions about the long-term advantage of accommodating resistance. The model also makes a falsifiable prediction that accommodating resistance should outperform fixed loading only over many sessions of progressive overload. This could also, in principle, be tested with a sufficiently long controlled study. Finally, our extension of the detraining term to scale with calendar time between sessions, rather than applying uniformly per session, adds realism that the original model lacks.

Future extensions of this work could address several of the limitations identified above. Most directly, future simulations should hold ω constant under progressive overload and increase only the attached fixed load component, which would better reflect realistic chain use and likely bring the chains result closer to Arandjelović’s predicted parity with bands. Making the fatigue time constant τ_f task-specific, following the framework of Enoka and Duchateau [3], would address the uniform decay rate assumption. More ambitiously, expanding the framework to multiple exercises by modeling individual muscle contributions rather than an aggregate net force would allow simulation of complete training programs and significantly increase the model’s practical utility for coaches and athletes.

Acknowledgments

We thank peer reviewers for their feedback on this project. One change we made based on conversations with them was revisiting several assumptions and clarifying aspects of the model that were either imprecisely stated or insufficiently motivated. This strengthened both the overall accuracy and clarity of our paper.

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